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Research paper

Classification of sp^2-bonded carbon allotropes in two dimensions

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ABSTRACT

A taxonomy for sp^2 carbon allotropes in two dimensions (2D) is proposed where single-layer structures are assigned a unique symbol associated with the geometry of each allotrope. The naming scheme is demonstrated for structures described in the literature and is further illustrated for a number of other topology-allowed sp^2 carbon systems. The symbol is easy to use and gives a direct access to geometrical features such as the number of polygons and their arrangement. It facilitates the classification of structures reported in the literature, where many such structures are found to have been assigned a name based on each author's somewhat arbitrary choice. This naming scheme will help rationalize future studies on this class of systems. A comparative study of the energetics of the structures is also performed, using a consistent set of numerical parameters. The naming scheme can be expanded to carbon systems with mixed hybridizations and to other non-carbon 2D systems.

Graphene research is a central subject in carbon nanoscience and the investigation of strategies to tune its physical properties remains a major focus of a significant part of the scientific community in chemistry, physics, and materials research [1]. Existing efforts include modification into quasi-1D nanoribbons and nanotubes [2,3], doping [4], and introduction of defects [5]. Defects can be studied in disordered [6] or highly-ordered [7] forms, and in variable concentrations. From the perspective of theoretical studies, extended distributions of defects have resulted in the investigation of 2D sp^2-bonded carbon lattices with geometries completely different from that of the honeycomb lattice [8,9]. The underlying principle is that one shifts the point of view where a graphitic structure is full of defects, to that of a new allotrope where defects are seen as basic, well-ordered building blocks. The different concepts behind the proposal of these non-graphitic structures include the extended distributions of bond-rotations [10], annealing or structural search methods [11], and condensation of molecular polycyclic precursors [12]. We found 45 such 2D sp^2-bonded carbon allotropes reported so far in the literature. With the recent advances in bottom-up synthesis [3], there is high optimism for the successful realization of many of these allotropes, as highlighted by the recent case of biphenylene [13].

The focus of this paper is to develop a unified approach to name those structures in a systematic and unambiguous way. We note that while other structures such as bilayer graphene can still be considered as two-dimensional systems, the naming scheme proposed here focuses on naming each monolayer of sp^2-bonded nanomaterials. The motivation stems from the fact that 2D sp^2-bonded carbon allotropes reported in the literature are named in many different ways, often based on the somewhat arbitrary choice of each author, and none of which being easily extendable to new nanostructures. In many cases, existing naming approaches are inspired by graphene seen as a parent structure. This includes systems such as graphenylene [14] and corographene [15]. In some structures, the naming can also be inspired by the polygons forming the system, such as in DHP-graphene [16], phagraphene [17], and QPHT-graphene [18]. Greek letters are also used to name some systems, *e.g.*, ψ-graphene [19], or series of structures, as in phographenes [20] and naphthylenes [21,22]. There are other examples where a given 2D s^2-bonded lattice is referred to by distinct names in studies from different authors, such as the system referred to as HOP-graphene [23], γ-phographene [20], and lipentoctahexatite [24]. There is also the case of the term biphenylene, which has been used to refer to two different structures, one with hexagonal [25] and another one with rectangular [13,26] symmetry. Clearly, identifying different allotropes in a precise way can be a difficult task.

1. Nomenclature

We propose a naming convention for sp^2-bonded two-dimensional crystal structures where all atoms are exactly threefold-connected. Even though this approach can be extended to structures made of other chemical elements, we focus solely on the case of carbon. We define

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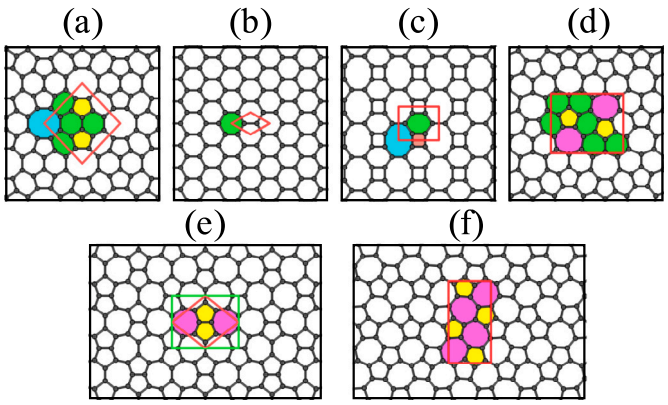


Fig. 1. Examples of sp^2 two-dimensional carbon nanostructures: (a) Cmmm-graphene ($c5^26^48^1$), (b) graphene ($h6^1$), (c) biphenylene ($r4^16^18^1$), (d) phagraphene ($r5^26^67^2$), (e) pentaheptite R57a ($c5^27^2$) and (f) pentaheptite R57b ($r5^47^4$). In each case, the primitive cell is highlighted in red. In the R57a case we also mark the conventional rectangular cell in green color.

a generic name of Λ-graphene with a prefix symbol given by

$$\Lambda = x\prod_n n^{\alpha_n} \tag{1}$$

where α_n is the number of polygons with n sides within the primitive cell of the system, and x identifies the geometry of the system's primitive cell. The resulting string lists the polygons in order of increasing number of sides they contain. Because there are only five different 2D crystalline systems, we have x = h, t, c, r, and o for hexagonal, tetragonal (or square), centered rectangular, rectangular, and oblique (monoclinic), respectively.

The easiest way to explain the notation is to examine a few examples. For instance, consider Fig. 1a where a system with 14 carbon atoms per primitive unit cell is shown. The unit cell corresponds to a centered lattice and features two pentagons, four hexagons, and one octagon, as highlighted by the polygons colored in the figure. It follows that the symbol is $c5^26^48^1$ and the structure is named $c5^26^48^1$-graphene. This structure corresponds to the system studied in Ref. [11]. Other structures shown in Fig. 1 can be named using the proposed scheme. For example, graphene (Fig. 1b) is classified as $h6^1$-graphene, (rectangular) biphenylene as $r4^16^18^1$-graphene (Fig. 1c), and phagraphene as $r5^26^67^2$-graphene (Fig. 1d). We stress that the nomenclature should be used relative to the system's primitive cell, in order to avoid erroneous classifications. For instance, let us consider the two pentaheptite structures shown in Fig. 1e–f, referred to as R57a and R57b in the literature. They both have four pentagon–heptagon pairs within a rectangular cell. However, the rectangular cell is primitive only for the R57b case ($r5^47^4$), while the correct symbol for R57a is $c5^27^2$, since it is a centered rectangular system.

Since each carbon atom in these sp^2 systems binds to exactly three other atoms, we can use Euler's rule to demonstrate that the symbol components are constrained to obey the known formula [27,28]

$$\sum_n (6-n)\cdot\alpha_n = 0. \tag{2}$$

This equation is based on Euler's rule and establishes that the average number of sides per polygon is exactly six. This is due to the fact that the Gauss curvature of the 2D system is zero, according to Gauss–Bonnet theorem and the fact that the 2D systems topologically correspond to toroidal structures. Based on these geometrical considerations, we can also obtain the number of atoms within the unit cell from the classification symbol:

$$N_{\text{atom}} = \frac{\sum_n n\cdot\alpha_n}{3} = 2\sum_n \alpha_n = 2N_{\text{poly}}, \tag{3}$$

where N_{poly} is the total number of polygons in the primitive unit cell.

For illustration purposes, we list 45 2D sp^2 carbon allotropes previously reported in the literature in Table 1, together with their names, the symbol proposed here, the wallpaper symmetry group, the number of atoms per primitive unit cell, lattice parameters and angles, total energy per atom (ΔE) relative to graphene, and reference of the publication where they previously appear. Note that, in the case of centered rectangular systems, we list the lattice data according to the primitive lattice vectors ($|\mathbf{a}_1| = |\mathbf{a}_2|$, $\theta \neq 60°, 90°, 120°$). Fig. 2 shows all the structures, with highlighted polygons and symmetry elements composing their corresponding wallpaper symmetry groups. Even though the main focus of this work is not to investigate relative energies for these systems in detail, it is interesting to note that among the 45 allotropes listed in Table 1, 30 of them have lower ΔE than biphenylene, a recently synthesized structure (see Appendix A for computational details on the density functional theory (DFT) calculations).

2. Uniqueness

All 45 allotropes studied in the literature are found to have a unique name according to the proposed scheme. We note that two different structures can have the same $\prod_n n^{\alpha_n}$ part of their symbols, but differ by their crystalline systems, *e.g.*, $r5^26^28^1$ and $c5^26^28^1$, and $o5^26^27^2$ and $r5^26^27^2$.

We highlight, however, that this uniqueness is not strictly true in general and we have identified cases where additional information is needed. This is illustrated in Fig. 3, where we show two distinct lattices with the same $o5^26^{24}7^2$ symbol. This kind of non-uniqueness is likely to occur for structures with large unit cells. In this example, the Dienes defect (also sometimes referred to as a Stone–Thrower–Wales reconstruction) [62] can be seen as a defect introduced in a graphene lattice rather than leading to a unique crystal. Such structures might continue to be named *graphene with a 5775 defect*. However, we recommend an approach where the defect is seen as a domain embedded in a larger structure. A description of the defects and the lattice vectors for their periodic images (*i.e.*, the lattice vectors of the supercell) in terms of graphene's primitive vectors, suffices to untangle the names. For instance, in the examples of Fig. 3a,b, we can write:

- $d5^27^2@h6^1$-[(4,4);(4,-3)] (Fig. 3a), and
- $d5^27^2@h6^1$-[(4,4);(5,-2)] (Fig. 3b).

Here, the 5775 defect is represented by $d5^27^2$ (with d standing for *defect*). The rest of the symbol indicates that the defects are periodically arranged at the $h6^1$ lattice points defined by the (4,4) and (4,-3) vectors in the case of the system from Fig. 3a, or by the (4,4) and (5,-2) vectors in the case of the system from Fig. 3b. Each (n, m) vector is defined in terms of $h6^1$-graphene's primitive vectors, $\mathbf{a}_1$ and $\mathbf{a}_2$ ($n\mathbf{a}_1+m\mathbf{a}_2$) in which the defect is embedded. Such a systematic description can be extended to any other carbon allotrope containing periodically arranged defects embedded in a larger structure.

In addition to the case of structures with a small density of defects embedded in a larger lattice, there are also rare examples of small-cell 2D crystals that are described by the same general formula. We did not identify any structure studied previously that falls in this category. However, to be exhaustive, we illustrate a case of non-uniqueness with the example in Fig. 4 that we found during a numerical search (see Appendix B). Other pieces of information, like the symmetry group, lattice constants, and/or lattice angles could be used to disentangle cases where a symbol corresponds to more than one structure. However, here we recommend extending the structure's symbol as:

$$\Lambda = x\prod_n n^{\alpha_n} - \{S_i\} \tag{4}$$

where the up to $N_{\text{poly}} \times \{S_i\}$ symbols are strings of characters (with the list of polygons neighboring the ith polygon) introduced for the

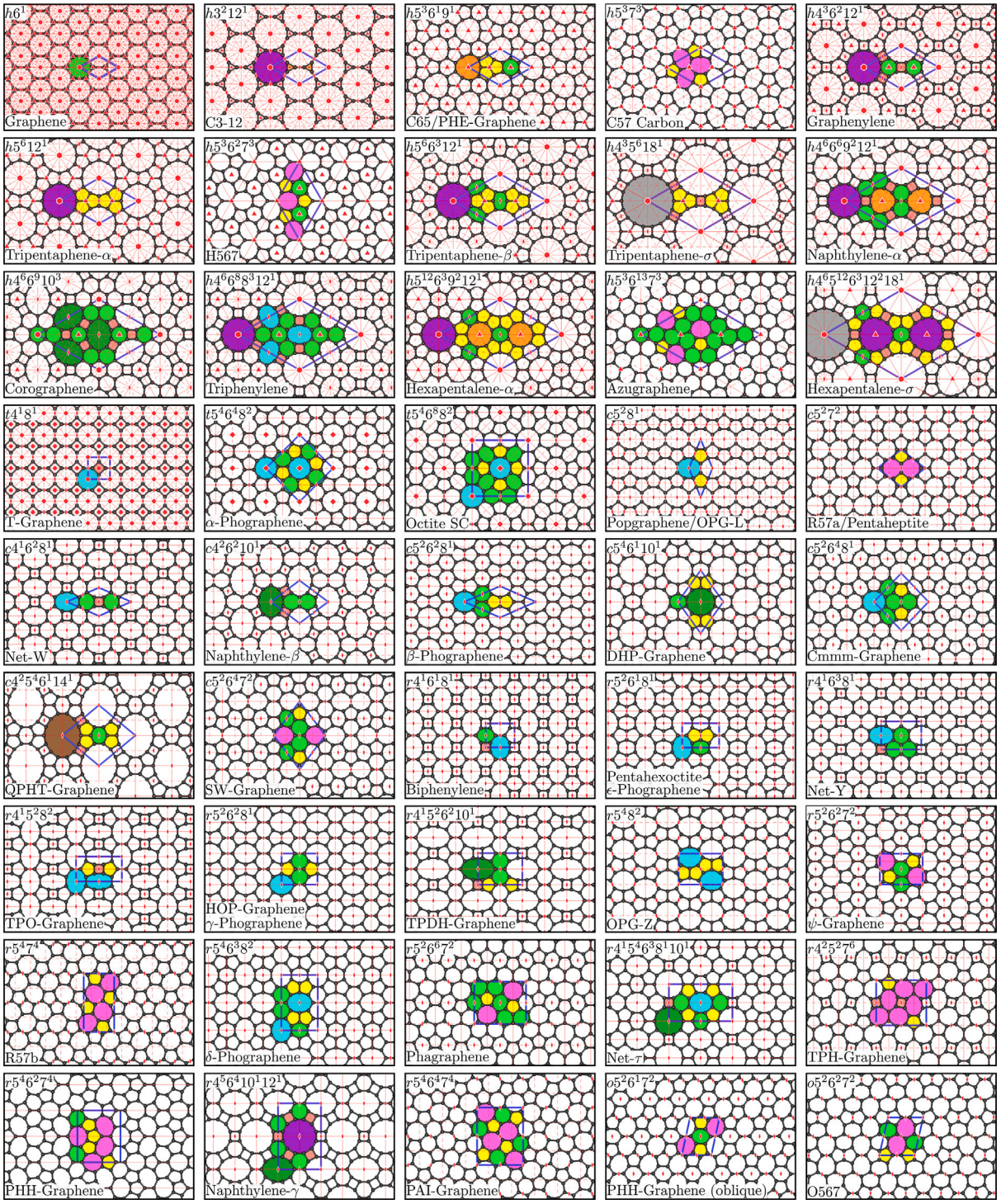


Fig. 2. Ball-and-stick representations of 45 sp^2 2D carbon nanostructures reported in the literature, along with the previously-used and proposed names. The proposed name is made clear by the highlighted polygons in each unit cell. The symmetry operations are shown in red. The CIF file of each structure can be downloaded at Ref. [29].

occasional cases where the basic formula is not sufficient. This symbol allows one to add information on the local environment of each polygon, starting from the smallest one, and adding more strings until a unique name is obtained. S_i strings are added until each label is unique. Once again, this is best explained in the example, as illustrated in Fig. 4. In the structure shown in Fig. 4a, one pentagon is neighbored by three hexagons and two octagons, so that its symbol is $c5^1_S5^16^28^1-\{S=6^38^2\}$. For the system in Fig. 4b, one pentagon is neighbored by another pentagon, two hexagons and two octagons, so that its symbol is $c5^1_S5^16^28^1-\{S=5^16^28^2\}$. In this case, the addition of one string is sufficient to disentangle the names. Clearly, in those cases, the notation symbol can become quite cumbersome but this fact is merely a translation of the complexity of the structure. More information regarding these structures are included in Table 2, including their wallpaper symmetry groups, lattice details, and the energy per atom relative to graphene.

Table 1
List of sp^2-bonded 2D carbons nanostructures reported in the literature. The previously introduced name, proposed symbol, wallpaper symmetry group, number of atoms in a primitive cell, DFT energy per atom (compared to $h6^1$-graphene), and lattice parameters and angles are provided. The lattice parameters and angles values are taken from DFT/PBE simulations described in Appendix A.

Structure	Symbol	Symmetry group	N_{atom}	a/b (Å)	θ	ΔE (eV)	References
Graphene	$h6^1$	$p6mm$	2	2.47/2.47	60°	0.000	[1]
C3–12	$h3^212^1$	$p6mm$	6	5.20/5.20	60°	0.966	[30]
C65/PHE-graphene	$h5^36^19^1$	$p3m1$	10	5.73/5.73	60°	0.401	[31–33]
C57	$h5^37^3$	$p31m$	12	6.15/6.15	60°	0.457	[34]
Graphenylene[a]	$h4^36^212^1$	$p6mm$	12	6.76/6.76	60°	0.622	[14,25,35–37]
TPH-α	$h5^612^1$	$p6mm$	14	7.15/7.15	60°	0.600	[33,38]
H567	$h5^36^27^3$	$p31m$	16	7.09/7.09	60°	0.233	[9,10]
TPH-β	$h5^66^312^1$	$p6mm$	20	8.50/8.50	60°	0.794	[38]
TPH-σ	$h4^35^618^1$	$p6mm$	20	9.58/9.58	60°	0.835	[33,38]
Naphthylene-α	$h4^66^92^12^1$	$p6mm$	30	10.30/10.30	60°	0.545	[21]
Corographene	$h4^66^910^3$	$p6mm$	36	11.06/11.06	60°	0.441	[15]
Triphenylene	$h4^66^83^12^1$	$p6mm$	36	11.07/11.07	60°	0.467	[39]
Hexapentalene-α	$h5^{12}6^39^212^1$	$p6mm$	36	11.09/11.09	60°	0.383	[40]
Azugraphene	$h5^36^{13}7^3$	$p31m$	38	10.84/10.84	60°	0.151	[41]
Hexapentalene-σ	$h4^65^{12}6^312^218^1$	$p6mm$	48	13.93/13.93	60°	0.645	[40]
T-graphene	$t4^18^1$	$p4mm$	4	3.45/3.45	90°	0.520	[30,42,43]
α-phographene	$t5^46^48^2$	$p4gm$	20	7.42/7.42	90°	0.191	[20,44,45]
Octite SC	$t5^46^88^2$	$p4mm$	28	8.77/8.77	90°	0.315	[46,47]
Popgraphene/OPG-L	$c5^28^1$	$cmm2$	6	4.90/4.90	44.31°	0.304	[48–50]
R57a/Pentaheptite	$c5^27^2$	$cmm2$	8	4.75/4.75	76.15°	0.239	[8–10]
Net-W	$c4^16^28^1$	$cmm2$	8	5.49/5.49	47.64°	0.355	[50,51]
Naphthylene-β	$c4^26^210^1$	$cmm2$	10	5.60/5.60	73.44°	0.531	[21]
β-phographene	$c5^26^28^1$	$cmm2$	10	6.09/6.09	47.61°	0.196	[20]
DHP-graphene	$c5^46^110^1$	$cmm2$	12	6.09/6.09	70.92°	0.365	[16]
Cmmm-graphene	$c5^26^48^1$	$cmm2$	14	6.18/6.18	87.14°	0.236	[11]
QPHT-graphene	$c4^25^46^114^1$	$cmm2$	16	7.42/7.42	79.53°	0.642	[18]
SW-graphene	$c5^26^47^2$	$cmm2$	16	6.68/6.68	74.26°	0.133	[52]
Biphenylene	$r4^16^18^1$	$pmm2$	6	4.53/3.76	90°	0.456	[13,26]
Pentahexoctite[b]	$r5^26^18^1$	$pmm2$	8	5.78/3.83	90°	0.350	[20,53]
Net-Y	$r4^16^38^1$	$pmm2$	10	6.28/4.39	90°	0.293	[54]
TPO-graphene	$r4^15^28^2$	$pmm2$	10	7.16/3.92	90°	0.663	[55]
HOP-graphene[c]	$r5^26^28^1$	$pmm2$	10	5.62/4.88	90°	0.237	[20,23,24]
TPDH-graphene	$r4^15^26^210^1$	$pmm2$	12	6.96/4.95	90°	0.445	[56]
OPG-Z	$r5^48^2$	$pmg2$	12	6.90/4.88	90°	0.353	[49]
ψ-graphene	$r5^26^27^2$	$pmg2$	12	6.71/4.84	90°	0.154	[19]
R57b	$r5^47^4$	$pgg2$	16	9.17/4.76	90°	0.216	[9,10]
δ-phographene	$r5^46^38^2$	$pmm2$	18	5.72/8.64	90°	0.258	[20]
Phagraphene	$r5^26^67^2$	$pmg2$	20	8.11/6.64	90°	0.193	[17]
Net-τ	$r4^15^46^38^110^1$	$pmm2$	20	9.96/5.77	90°	0.331	[57]
TPH-graphene	$r4^25^27^6$	$pmg2$	20	7.84/7.09	90°	0.386	[12,58]
PHH-graphene	$r5^46^27^4$	pm	20	6.74/8.11	90°	0.315	[59]
Naphthylene-γ	$r4^56^410^112^1$	$pmm2$	22	6.73/10.35	90°	0.576	[22]
PAI-graphene	$r5^46^47^4$	$pgg2$	24	7.20/9.00	90°	0.134	[60]
PHH-graphene[d]	$o5^26^17^2$	$p2$	10	4.63/6.02	77.69°	0.244	[59]
O567	$o5^26^27^2$	$p2$	12	5.52/6.09	77.01°	0.357	[61]

[a]Also coined as ϵ-phographene.
[b]Also called biphenylene carbon (BPC).
[c]Also referred to as γ-phographene or lipentoctahexatite.
[d]Oblique geometry version of PHH-graphene.

Table 2
Lattice details for the systems illustrated in Figs. 4 and 5, including the proposed symbol, wallpaper symmetry group, lattice parameters and angles, as well as the energy per atom ΔE relative to graphene.

Structure symbol	Sym. group	a/b (Å)	θ	ΔE (eV)
$c5^1_S5^16^28^1-\{S=6^38^2\}$	$cmm2$	7.30/7.30	31.11°	0.338
$c5^1_S5^16^28^1-\{S=5^16^28^2\}$	$cmm2$	6.09/6.09	47.61°	0.196
$r4^28^2$	$pmm2$	3.88/5.89	90°	0.844
$o4^15^16^19^1$	$p1$	5.50/4.41	78.10°	0.880
$c4^15^17^18^1$	cm	6.14/6.14	36.78°	0.600
$c4^16^28^1$	$cmm2$	5.49/5.49	47.64°	0.355
$r5^26^18^1$	$pmm2$	3.83/5.78	90°	0.350
$c5^27^2$	$cmm2$	4.75/4.75	76.15°	0.239
$h4^312^1$	$p3m1$	4.95/4.95	60°	1.488
$r4^26^110^1$	$pmm2$	4.56/5.17	90°	0.791
$o4^15^210^1$	$p2$	5.00/5.04	72.15°	0.611

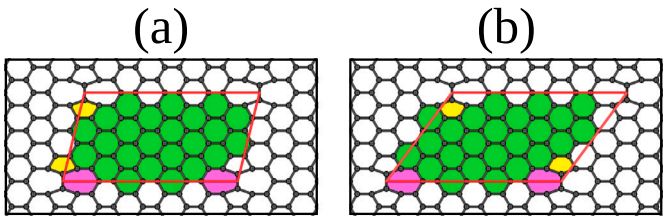


Fig. 3. Examples of two distinct sp^2 lattices with the same $o5^26^{24}7^2$ symbol. We classify these structures as (a) $d5^27^2@h6^1$-[(4,4);(4,-3)] and (b) $d5^27^2@h6^1$-[(4,4);(5,-2)].

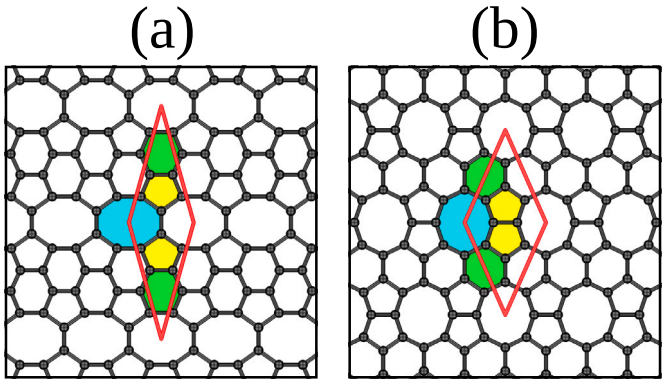


Fig. 4. Two different structures with the same basic $c5^26^28^1$ symbol. The symbol is extended by selecting the smallest polygon (here a pentagon) and listing all its neighboring rings as a complementary string. The (a) structure is thus uniquely referred to as $c5^1_S5^16^28^1-\{S=6^38^2\}$ while structure (b) is labeled as $c5^1_S5^16^28^1-\{S=5^16^28^2\}$.

3. Applications

To further illustrate the proposed naming scheme, we considered all the primitive cells that can be obtained by placing 8 threefold-connected atoms (4 polygons) in a 2D crystal (see Fig. 5). Most of the structures found here have not been studied before. Note that our search algorithm described in Appendix B yielded a number of structures featuring triangles. These are not rejected topologically but are unlikely to occur in organic chemistry and we do not show them in the figure. Conversely, squares are not frequently observed but they are possible [63], so those systems are included in the $N_{\text{atom}} = 8$ tiling configurations shown in Fig. 5. For 4 polygons, the naming provides a unique description for each structure. In general no additional S labels are required except in rare circumstances. We also provide computed data on lattice and energetics for all these structures in Table 2.

4. Conclusions

We have proposed a systematic naming scheme for 2D sp^2-bonded nanostructures with generic name Λ–graphene where $\Lambda = x\prod_n n^{\alpha_n} - \{S_i\}$. This symbol offers a clear standard to uniquely describe two-dimensional sp^2 monolayer carbon nanostructures. This notation is also a metasymbol, providing straightforward information on polygon count. The use of the symbol will ease the identification and the tracking of theoretically proposed carbon allotropes in the literature. In addition to the 45 structures already reported in the literature, there are many additional ways to periodically tile a plane using polygons with 3 edges meeting at each atomic site, as we have illustrated for the $N_{\text{atom}} = 8$ systems. The naming scheme can be expanded to 1D and other 2D non-carbon nanostructures as well (See Appendix C). The energetics and structural properties of all systems analyzed here have been determined within DFT/PBE. The data indicate that a number of structures show theoretical stability at least as high as that of some of already synthesized structures. All the structures discussed here are available for download in CIF format at Ref. [29].

CRediT authorship contribution statement

Eduardo Costa Girão: Conceptualization, Methodology, Writing – original draft, Editing. **Alastair Macmillan:** Software, Reviewing, and editing. **Vincent Meunier:** Methodology, Writing – original draft, Editing, Data curation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Energetics calculation

The relative 0 K stability energy values reported here were all calculated within DFT using the GPAW [64,65] package with analysis tools provided by the atomic simulation environment (ASE) package [66]. GPAW is a python-based code package which utilizes the grid-based projector augmented wave (PAW) method to describe electron-ion interactions [67]. We employed the Perdew–Burke–Ernzerhof (PBE) approximation to describe exchange–correlation [68]. A distance of at least 15 Å was added in the (z) direction perpendicular to the plane of each structure to avoid any issue with interaction between images in that direction. An energy cut-off of 600 eV was found to yield numerically converged data. The lattice parameters and atomic configurations were relaxed simultaneously with a quasi-Newton method until the forces were lower than 50 meV/Å. With the notable exception of reciprocal space sampling, all the other parameters were kept the same for all calculations. For k-point sampling, we first determined that a $9 \times 9 \times 1$ Monkhorst–Pack (MP) k-point grid sampled $h6^1$-graphene Brillouin zone with sufficient accuracy to obtain numerically converged results. For all the other structures, we determined a lattice-specific $n_1 \times n_2 \times 1$ MP sampling that corresponds to a density of k-points that is at least as high as that used for $h6^1$-graphene in each direction.

Appendix B. A numerical algorithm for threefold connected tiling

We now describe the search algorithm we developed to find tiling solutions, such as the structures shown in Fig. 5. The constraints of our search are (1) the structures are 2D periodic and (2) each atom is exactly threefold connected (*i.e.*, to three different atoms). Euler's equation (Eq. (2)) and its corollary establish the necessary conditions a structure must meet in order to satisfy those constraints. We shall show below that Euler's condition is not sufficient for the prediction of the type of planar tiling we are interested in. Eq. (3) imposes that only systems with an even number of atoms can yield a threefold coordinated structure in 2D. The starting point of our recursive algorithm hinges on that property.

Stage one — calculating connectivity matrix: The first stage of the algorithm proceeds as described in Fig. 6. Starting from the vacuum state ($N = 0$, where N is the number of polygons), the addition of 2 atoms (a dimer) can only yield one type of threefold coordinated system: $x6^1$-graphene ($N = 1$). The next step, $N = 2$, consists in adding one dimer to the single state corresponding to $N = 1$. This is illustrated in Fig. 6. By inspection, we conclude that adding a dimer can only be done in three non-equivalent ways, yielding $x3^19^1$, $x4^18^1$, and $x6^2$ (exhaustively). As the first stage of the algorithm operates on the connectivity matrix rather than the coordinates themselves, x does not need to be defined at this stage. The algorithm continues to $N = 3$ by

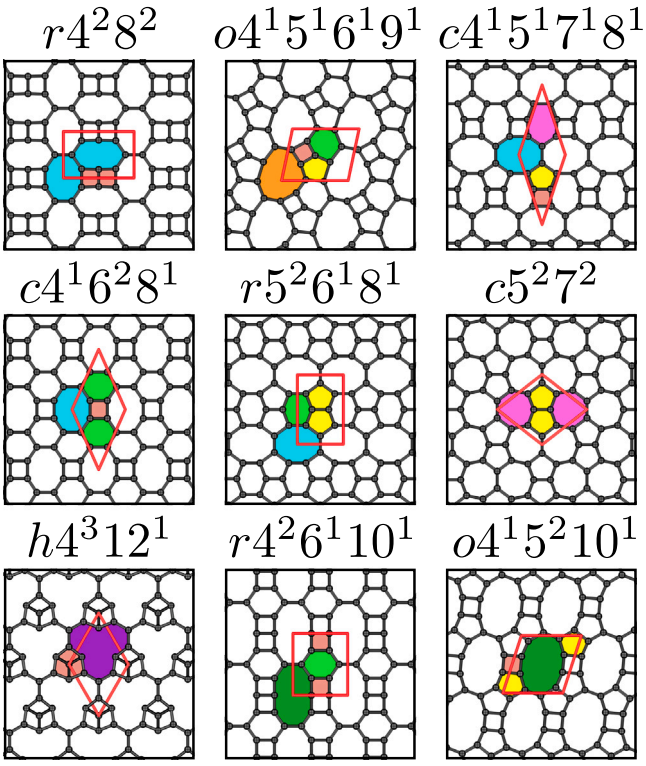


Fig. 5. Nine different structures containing 8 atoms within the primitive cell. These are all the possible ways to arrange eight threefold-coordinated atoms in a 2D periodic structure. Note that cells featuring triangles and non-primitive cells (*e.g.*, graphene) are not included. For instance, there is another structure featuring two squares and two octagons but that structure can be reduced to a 4 atoms per-cell primitive system (*i.e.*, $t4^1 8^1$-graphene, see Table 1). Note that $h4^3 12^1$-graphene is non-planar as corrugation in the out-of-plane direction is needed to ensure proper connectivity. The CIF file of each structure can be downloaded at Ref. [29].

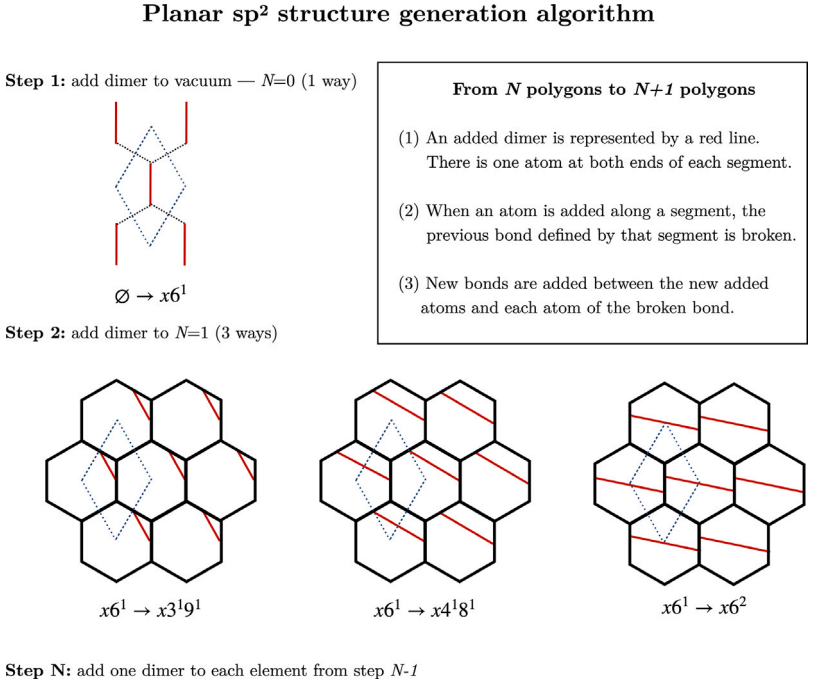


Fig. 6. Fundamental steps of a search algorithm for structures that are periodic in two dimensions and in which all atoms are threefold-coordinated. The algorithm is recursive and starts from the vacuum state with each step corresponding to the addition of a dimer, as prescribed by Euler's condition. In practice, the algorithm proceeds in two stages. First, the connectivity table of the structure is modified based on the steps shown in the figure. Second, Cartesian coordinates are obtained by relaxing the structure under the constraint of fixed connectivity matrix.

adding dimers to the three structures obtained for $N = 2$ and so forth for $N = 4, \ldots$ We note that since each dimer addition is a controlled modification of the connectivity matrix, one can easily keep track of what polygons are added and removed; thus making it possible to track the name of each tiling solution during the generation process.

Stage two — calculating coordinates: The procedure we describe here provides a handle on the connectivity matrix of each system, which we treat as a non-directed graph. In the next stage of the process, we evaluate the Cartesian coordinates of the discovered structures by performing a structural relaxation under the constraint of fixed connectivity. This can be done by introducing forces with simple harmonic contributions in terms of bond angles and distances for the nearest-neighbors [69]. We found that adding a weak repulsive force between second nearest-neighbors helps to avoid structural collapse. The resulting structures can then be relaxed using more advanced methods such as density functional theory. This last step allows the determination of the x value for each lattice.

Euler criterion is not a sufficient condition: This algorithm also demonstrates the reason why many structures that would meet Euler's criterion do not mathematically exist. This is best demonstrated by the absence of $x5^17^1$ for $N = 2$. The fundamental reason for this absence is related to dislocation theory and the fact that a single 57 pair in a threefold-coordinated network is a structural dislocation with a net non-zero Burgers vector. As a result, it cannot, alone, constitute a periodic threefold coordinated crystal [70,71]. It follows that $x5^16^17^1$ does not exist either, since it could be transformed into $x5^17^1$ by a dimer removal operation. In contrast, $x5^27^2$ is allowed because, in that case, the 57 pairs are aligned in such a way as to make the sum of the Burgers vectors zero within each unit cell. The absence of $x5^17^1$ demonstrates that Euler's criterion is a necessary but not sufficient condition to predict the type of tiling. As N increases, a large number of structures (that could be reduced to $x5^17^1$ by dimer removal) that would meet Euler's criterion are not appropriate tiling solutions under the threefold coordination and two-dimensional constraints.

To be complete, one should note that the algorithm does not generate all possible tiling solutions without the inclusion of one additional operation. For instance, somewhat exotic structures featuring abutting triangles such as $x3^212^1$ do not reduce to the vacuum state by sequential removals of dimers while preserving threefold connectivity at each step. Instead, it reduces to disconnected graphs. However, adding a procedure of bond rotation to the algorithm fixes this issue. For example, $x3^212^1$ can be obtained by an additional transformation consisting in a bond rotation in structure $x3^14^111^1$, which itself has $x3^19^1$ as a direct parent and $x6^1$ as grand-parent.

Appendix C. Systems with mixed hybridizations

Even though the proposed symbol is motivated by studies on full sp^2 systems, it can be also extended to mixed $sp^2 - sp^3$ quasi-2D carbon nanostructures, like pentagraphene ($t5^4$, see Fig. 7a) [72], tetrahexcarbon ($c4^26^2$, Fig. 7b) [73,74], and PH-graphene ($h5^{12}6^1$, Fig. 7c) [75]. However, one may note that Eqs. (2) and (3) are no longer valid, since these structures feature sets of tetra-coordinated atoms.

The situation is more challenging for $sp - sp^2$ graphyne (GY) structures, since acetylenic bonds lie along the edges of the structures polygons. GYs were first proposed by Baughman and Eckhardt (BE) in 1987 [76], where the authors presented an initial method for naming these structures. This was based on the number of carbon atoms in the smallest ring, followed by the second smallest ring, and so on, where subsequent rings are constrained to be linked by a C^{sp^2}–$C^{sp} \equiv C^{sp}$–C^{sp^2} rod. One fundamental difference to our sp^2 proposal is that two images of a given ring (related to each other by a Bravais lattice translation) can appear in the BE naming approach. In addition, BE naming does not necessarily give information of all pores within a unit cell. However, special examples of GY systems are usually simply referred to by Greek letters like α-, β-, δ-, and γ-GY [77–79].

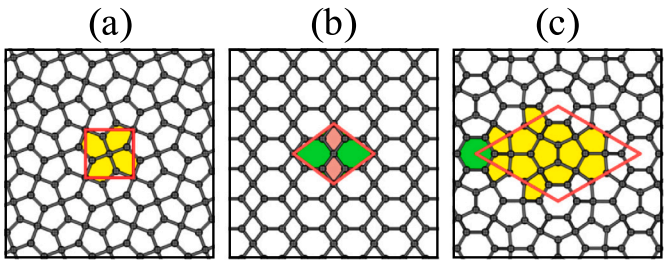


Fig. 7. Examples of systems with both sp^2 and sp^3 atoms: (a) pentagraphene ($t5^4$), (b) tetrahexcarbon ($c4^26^2$), and (c) PH-graphene ($h5^{12}6^1$).

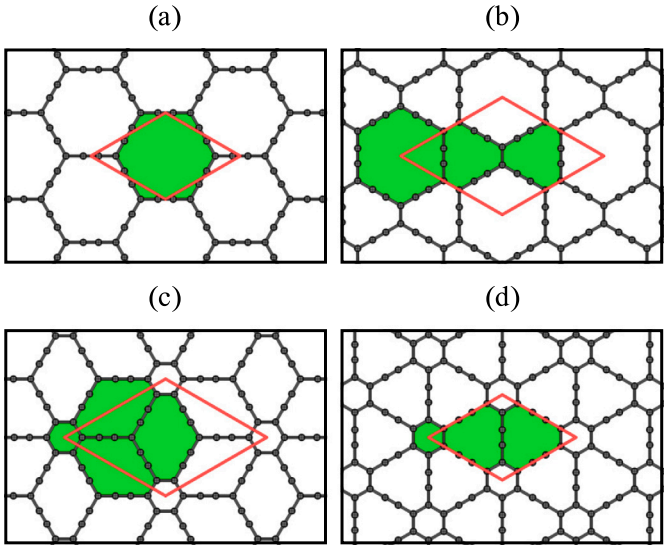


Fig. 8. Examples of graphyne (GY) lattices and their corresponding symbols. (a) α-GY ($h6^1_6$), (b) β-GY ($h6^2_36^1_6$), (c) δ-GY ($h6^1_06^3_4$), and (d) γ-GY ($h6^1_06^2_3$).

GY-like structures can assume a broad set of forms. Serafini et al. have explored many of these structural variations [80] and they used a naming scheme based on special units like **h**, **H**, **R**, **T**, **L** (standing for small and large hexagons, rhombus, triangle, and line, respectively), as previously proposed by Park, Kim, and Lee (PKL) [81]. While the PKL approach has similarities to ours, it is limited to five types of GY pores (**h**, **H**, **R**, **T**, **L**), corresponding to 6-membered polygons (considering either a $C^{sp^2} - C^{sp^2}$ or a $C^{sp^2} - C^{sp} \equiv C^{sp} - C^{sp^2}$ unit as single sides). In this way, it is not trivial to extend it to describe non-conventional GYs involving, for example, 4-, 5-, 7-membered pores, and beyond [82–85].

If we just extend the use of our symbol to GYs, α-GY (Fig. 8a) would be classified as $h6^1$, while the β- and γ-GY (Fig. 8b,d, respectively) cases would both be classified as $h6^3$, with these two last cases differing by the distribution of $C \equiv C$ bonds. We highlight, once again, that here we treat both $C^{sp^2} - C^{sp^2}$ and $C^{sp^2} - C^{sp} \equiv C^{sp} - C^{sp^2}$ sectors as single sides of a given polygonal pore. In order to distinguish systems like β- and γ-GY, we propose to expand our classification symbol by differing rings with the same number of sides by the number of acetylenic bonds within them. It can be written as

$$\Lambda = x \prod_n n^{\alpha_n}_{\beta_n} n'^{\alpha_{n'}}_{\beta_{n'}} n''^{\alpha_{n''}}_{\beta_{n''}} \ldots \tag{C.1}$$

for the *Λ-graphyne* structure, where n, n', and n'' correspond to polygons with the same number of sides, but are associated to rings with β_n, $\beta_{n'}$, and $\beta_{n''}$ acetylenic bonds. In this sense, α-GY has a single hexagon in its primitive cell (with six acetylenic bonds), so that the system would be called as $h6^1_6$-graphyne. The symbol for β-GY is $h6^2_36^1_6$, since its unit cell contains two hexagons (one hexagon) with 3 (6) acetylenic bonds, while δ-GY (Fig. 8c) is labeled as $h6^1_06^3_4$ since it has one hexagon (two

hexagons) with no (4) acetylenic bonds in the unit cell. Finally, γ-GY is the $h6_0^16_3^2$-graphyne structure since it has one hexagon (two hexagons) with no (3) acetylenic bonds in the unit cell.

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